



Extreme Performance and Efficient Application

A6-RS is a new-generation high-performance servo drive system promoted by STEPPERONLINE. This system is dedicated to realizing optimized configurations with higher productivity and production efficiency and lower production costs and providing drive solutions with optimal balance between performance and costs, helping your business success.

The A6-RS system consists of the series A6-RS servo drive and the A6M60 or A6M80 series servo motor, with the power in the range of 0.2kW to 1.0kW. The pulse reference mode is adopted, meeting various standard applications.

1.1 Features



Excellent performance:

- High response: 2 kHz speed loop bandwidth
- High accuracy: 17-bit absolute encoder
- High efficiency: highly dynamic and high precision to achieve higher productivity



Reliable operation:

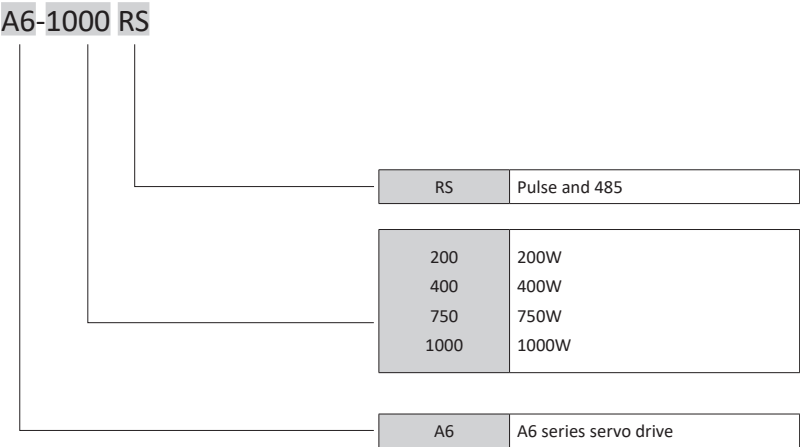
- Built-in dynamic brake function.
- High-quality motor bearings extend the service life.
- The drive can run stably in harsh environment.
- The product is CE certified, meeting the equipment export requirement.



Easy-to-use:

- Compact design, making the product be able to be installed in narrow space.
- Easy to connect and use. The USB cable is used to improve debugging efficiency.

1.2 Model



1.3 Nameplate

Nameplate

Model →

Rated input →

Rated output →

Manufacturing SN →

STEPPERONLINE

MODEL: AS-400RS

INPUT: 1PH AC200~240V 4.0A 50/60Hz

OUTPUT: 3PH AC0~240V 2.8A 0~500Hz 400W

S/N: CA000445220400001



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1.4 Components

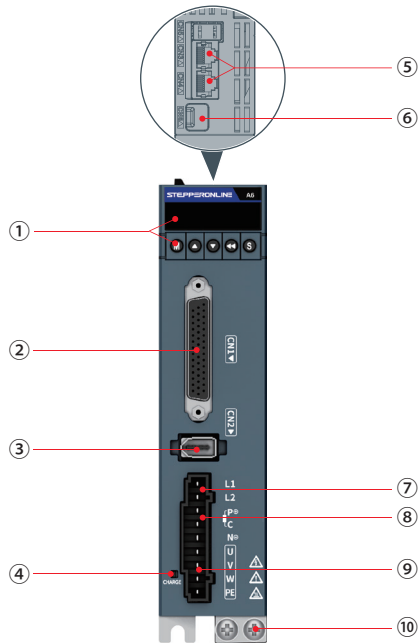


Figure 1-1 Components of the A6-RS servo drive (SIZE A)

No.	Name	No.	Name
①	Display and operation area	⑥	Commissioning and communication port (CN6)
②	Control signal port (CN1)	⑦	Power input
③	Encoder signal port (CN2)	⑧	Braking resistor port
④	Charging indicator	⑨	Motor power output
⑤	Communication port (CN3 and CN4)	⑩	System ground

NOTICE

- The above figure describes the component layout of the SIZE A drive. Component layout of other models may be different. For positions of ports of other models, see section 3.3 Ports.

1.5 Rated Data

Single-phase 220 V servo drive

Item	SIZE-A		SIZE-B
Power	0.2kW	0.4 kW	0.75 kW
Drive model	A6-200RS	A6-400RS	A6-750RS
Continuous output current (Arms)	1.6	2.8	5.5
Maximum output current (Arms)	5.8	10.1	16.9
Main circuit power supply	Single-phase 200–240 V AC, –10% to +10%, 50/60 Hz		
Control circuit power supply	Powered up by the bus, sharing one power supply and rectification part with the main circuit		
Braking capability	External braking resistor		Built-in braking resistor

Single-phase/Three-phase 220 V servo drive

Item	SIZE-C
Power	1.0 kW
Drive model	A6-1000RS
Continuous output current (Arms)	7.6
Maximum output current (Arms)	23
Main circuit power supply	Single-phase/three-phase 200–240 V AC, –10% to +10%, 50/60 Hz
Control circuit power supply	Single-phase 200–240 V AC, –10% to +10%, 50/60 Hz
Braking capability	Built-in braking resistor

1.6 Technical Specifications

Basic specifications

Item	Specifications
Control mode	IGBT PWM control, sine wave current drive mode 220 V, 380 V: Single-phase or three-phase full-bridge rectification
Encoder feedback	23-bit / 26-bit multi-turn absolute encoder, which can be used as a single-turn absolute encoder in absence of the battery
Operating temperature	0°C to +55°C (over 45°C: derate 10% for every additional 5°C)
Storage temperature	−40°C to +70°C
Altitude	Up to 2000 m. For altitude above 1000 m, derate 1% for every additional 100 m
IP rating	IP20 (except for the power terminal IP00)

Speed/torque control mode

Item	Specifications
Speed control range	1:6000 (The lower limit is the threshold within which the servo drive keeps running with the rated torque load.)
Frequency characteristic	2 kHz
Torque control accuracy	±1%
Soft start time setting	0 s to 3600 s (This parameter can be set for acceleration and deceleration separately.)

Position control mode

Item	Specifications
Feedforward compensation	0% to 100.0% (A resolution of 0.1% FS)
Input pulse form	Including "direction+pulse", "Phase A + B quadrature pulse" and "CW/CCW pulse reference form"
Input pulse frequency	Differential input: Up to 4 Mpps at high speed, and the pulse width cannot be less than 0.125 μs Open collector: Up to 200 kpps, and the pulse width cannot be less than 2.5 μs
Multi-position reference	Position 0 to position 15 selectable through DI signal combination
Output form	Phase A, phase B: differential output; Phase Z: differential output or open collector output

Built-in functions

Item	Specifications
Overtravel (OT) prevention	The servo drive stops immediately when P-OT or N-OT becomes active
Protection functions	Protections against overcurrent, overvoltage, undervoltage, overload, main circuit detection error, heatsink overheat, overspeed, encoder error, CPU error, and parameter error
LED display	Main power supply CHARGE indicator, 5-digit LED display
Vibration suppression	Five notches (including two adaptive notches), 50 Hz to 8000 Hz
Connection protocol	USB
Communication protocol	Modbus
Multi-station communication	Communication: When RS485 is used, up to 128 nodes can be connected
Axis address setting	Based on the user parameter setting
Function	Status display, user parameter setting, monitored value display, alarm tracing display, jogging, and speed/torque reference signal observation
Other	Gain tuning, alarm record, I/O setting, and jogging